



Enhancing Triage Accuracy &
Efficiency Through AI

OPTISENSE TRIAGE

Transforming Emergency Care with AI — delivering faster, more accurate triage decisions, better patient outcomes, and smarter resource allocation. Together, we can revolutionize emergency department efficiency and save lives.



Problem Statement

High-Pressure Environment

01

Emergency departments operate under extreme time-critical conditions where every second counts. Staff face constant pressure to make rapid decisions with limited information.

Inconsistent Manual Triage

02

Manual triage processes lead to inconsistent decisions across different nurses and shifts. Two critical failure modes emerge: under-triage of high-risk patients and over-triage of stable patients.

30-40% Misclassification

03

Studies show misclassification rates of 30-40% in traditional triage. This leads to delayed care for critical patients, strained resources, and increased mortality rates.

Solution Overview

01

Emergency departments lose critical minutes every time a nurse manually reviews vitals and makes a triage call under pressure — OptiSense Triage eliminates that bottleneck by delivering an instant, AI-generated severity score the moment a patient walks in.

02

The system listens to what the patient says, reads their vitals, and cross-references their medical history to decide in seconds whether they need immediate resuscitation or can safely wait.

03

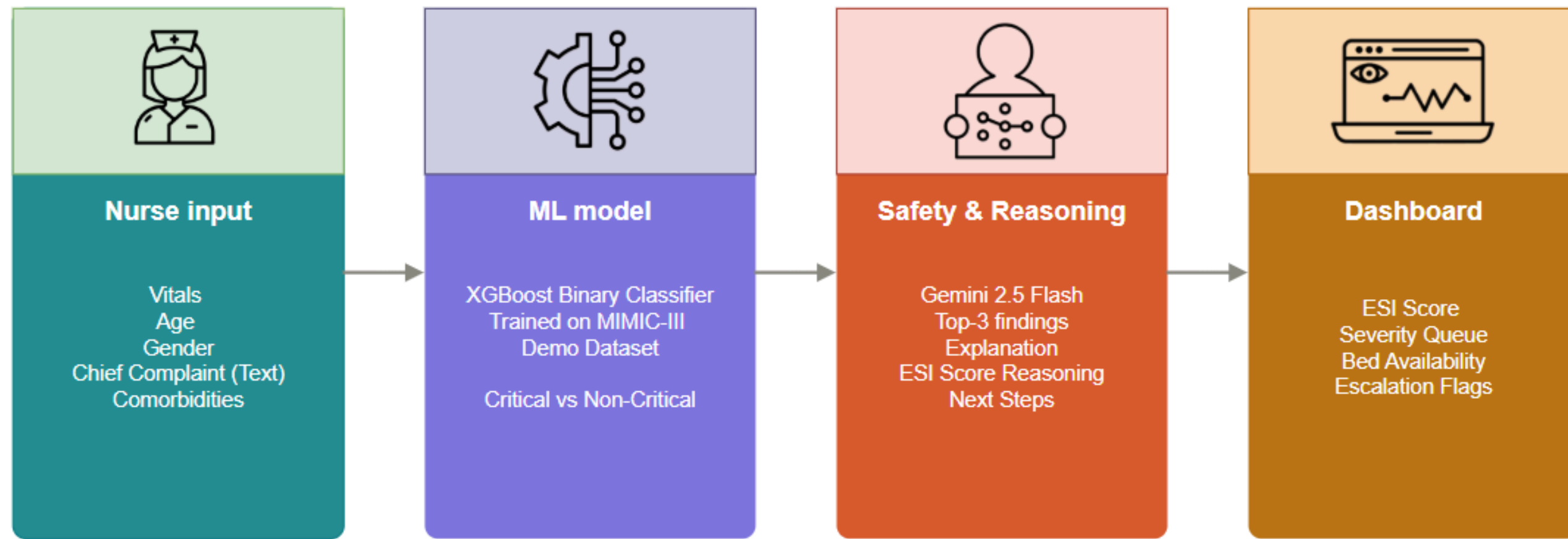
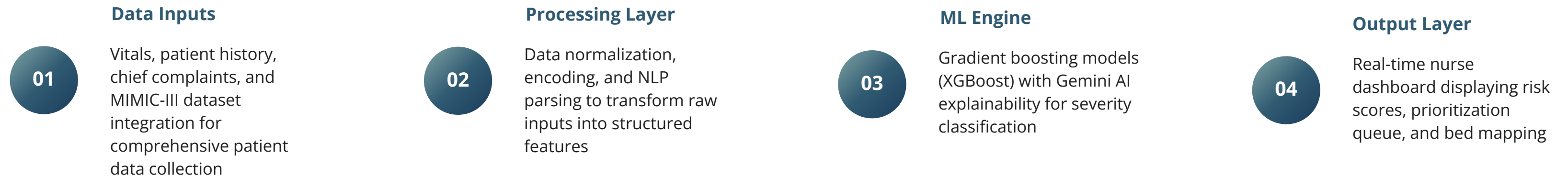
Every decision comes with a plain-English clinical summary and a prioritised action plan, so the attending nurse knows not just what to do but why and in what order.

04

The full patient queue is visible in real time, sorted by severity, with bed availability updated live.



Technical Architecture



CLIENT SIDE

Nurse Dashboard (Patient Queue, Beds)
React · Vite · Tailwind CSS

API GATEWAY

FastAPI
POST /triage · GET /queue · GET /beds · GET /health

AI & PROCESSING

Triage model
XGBoost Binary · Calibrated

Safety override
Clinical rules · ESI-1/2/3/4/5

Signal engine
Vitals · Shock Index · Review Flag

Gemini 2.5 Flash
Findings · Next Steps · Fallback

DATA LAYER

MIMIC-III
Training data source

Model artifacts
.pkl files · PCA · medians

Complaint embeddings
all-MiniLM → PCA(8)



Market Relevance

01

Global ED Crisis

Emergency departments face rising patient volumes, staffing shortages, and time pressure, while studies show manual triage misclassification rates of up to 30–40%, highlighting a clear and costly problem this system addresses.

02

Rapid Market Growth

The global market for AI-driven triage and clinical decision support systems is rapidly expanding, with projections showing multi-billion-dollar growth over the next decade, indicating strong commercial demand for solutions like this.

03

Alignment with Healthcare AI Adoption

Healthcare providers are increasingly adopting AI tools that support, rather than replace, clinicians, making a real-time “second opinion” triage system well-aligned with current adoption trends.

04

Target Markets

Primary users include hospitals, urgent care centers, and government healthcare systems seeking to optimize triage accuracy and resource allocation.

Scalability Potential

01

Model improves with more hospital data through network effects, enabling continuous learning and enhanced accuracy across healthcare systems.

02

Expand to ICU admission prediction, helping clinicians identify patients at high risk of requiring intensive care before deterioration occurs.

03

Early Detection capabilities to identify life-threatening infections before they become critical, improving survival rates significantly.

04

Beyond the ED, the same triage engine can extend to urgent care centres, ambulance dispatch systems, and remote triage hotlines — anywhere a fast, consistent severity decision has clinical value



Future Roadmap

-  Real-time EHR Integration
-  Multilingual NLP for Global Use
-  Regulatory Compliance (HIPAA/FDA/CE)
-  Peadiatric and trauma-specific models
-  Predictive Deterioration Alerts
-  IRB and Clinical Validation



OptiSense Triage

THANK YOU